Usability testing

What is usability testing?

Why would we use usability testing?

Usability testing

- **Usability testing** is a technique to evaluate the usability of an interface by testing it on users

- **Important:** users actually ***use*** the interface during a usability test
  - Users are typically asked to carry out tasks that the interface performs
  - Tester(s) will observe the user using the interface, while listening and taking notes

- Usability testing can be used for verification of usability-related requirements

- Usability testing can ***also*** be used as part of user analysis to better understand users, requirements, etc.
  - i.e. we can make revisions to requirements, software, interface design, etc, in resposne to usability test results

Why conduct usability testing?

- Identify usability issues
  - Which helps to identify changes required to improve usability (performance, subjective satisfaction, etc.)
- Learn if participants can successfully complete tasks and how well they can complete them
  - e.g. How long will it take them to complete tasks?
  - We can record metrics for performance (e.g. time to complete tasks), error rates and other things of interest
  - Can then use these metrics to verify if requirements are met
- Find out how participants think and feel about the interface, e.g. how satisfied are participants with it?



How do we conduct usability tests?

Source: https://cdn.business2community.com/wp-content/uploads/2016/09/usability-lab-3-2.jpg

Conducting usability tests

- Formal usability testing labs are not required!
- Many variations on how to conduct usability testing
  - We'll break down different forms of usability testing and tools we can use as part of the process
- Usability testing can also take place at different stages of development
- Fundamentally... if you're asking a user to perform tasks your application accomplishes while observing their behavior, you're usability testing

Conducting usability tests

- Good usability tests will have **test plan** that documents all details for how testing will occur
  - Test plans not one-size-fits-all, can be tuned to the interface, users and learning goals for the testing
- Participants will need to be **recruited** to participate in the usability test
  - How many participants? Depends on goals for testing.
- Findings will need to be **analyzed and reported**
  - Reports can have varying degrees of detail and format

Types of usability tests

Sketch, wireframe and mockup usability tests

- User is asked to interact with sketches and other non-interactive early-stage prototypes

- Tester flips pieces of paper or manually changes the screen in response to user actions
  - Fun
    example: https://www.youtube.com/watch?v=9wQkLthhHKA

- Fascinatingly... users have been shown to provide more open feedback to low-fidelity prototypes!
  - Why? Low fidelity implies low effort thus far, compared to telling the test the full-blown application is no good!

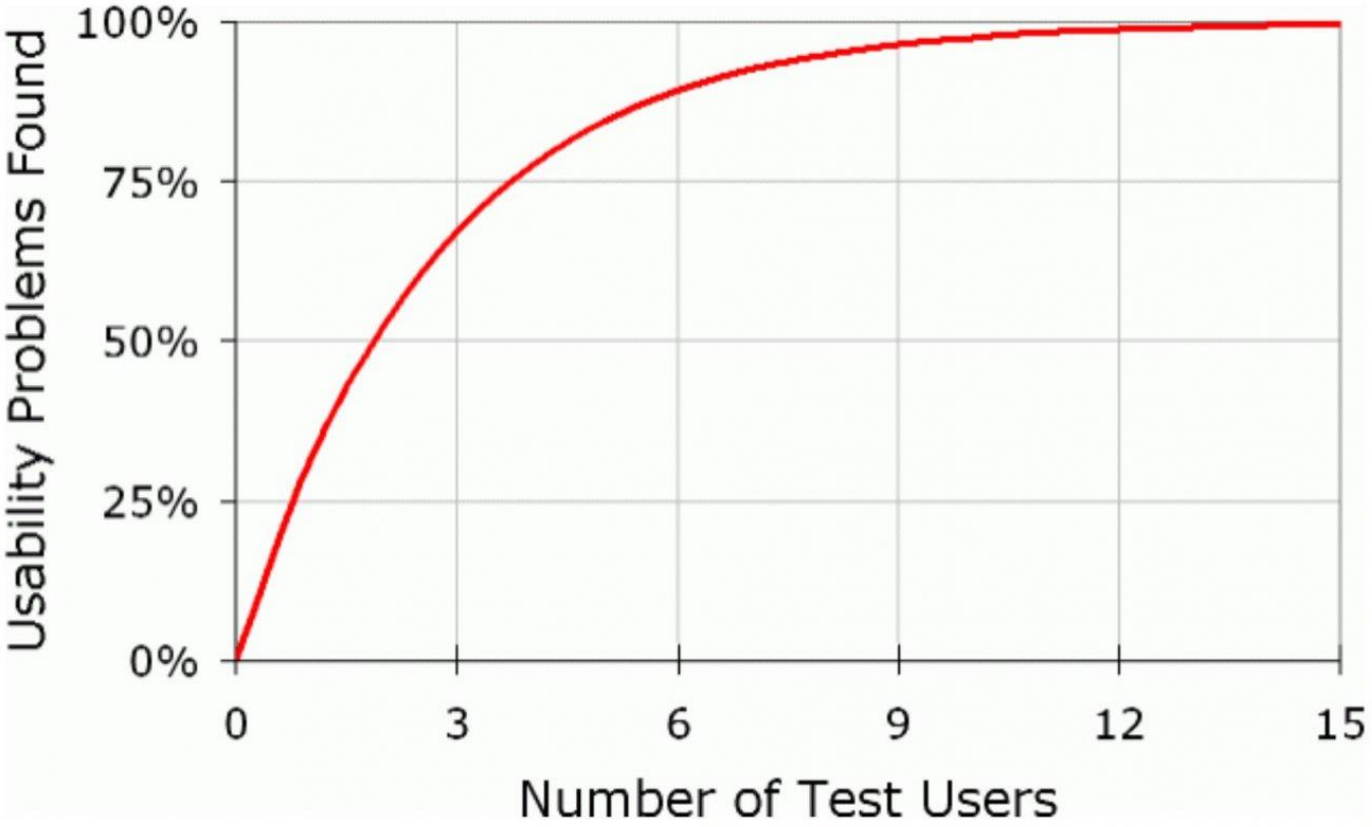
Discount usability testing

- More focus on qualitative observations, earlier stage prototypes, use of the think-aloud method, and testing with about 5 participants

- Qualitative observations require less time to prepare
  - Compared to more quantitative methods like surveys and collecting other data

- **Think-aloud techniques** involve asking participants what they are thinking as they use the interface

Testing with only 5 users? Yes!



Source: https://www.centigrade.de/wordpress/wp-content/uploads/nielsen-usability-test-with-5-users.jpg

Testing with only 5 users

- Research has shown it's actually enough to find a majority of usability issues
  - A lot of time spent with additional users is simply identifying the same issues over and over
- Critics do exist and are correct to point out that this will leave usability issues unidentified
- However low number of test participants encourages ***repeated*** testing
  - Imagine you have 25 participants worth of testing to "spend", is it better to use 25 on one iteration of the interface, or 5 for several iterations of testing?

Testing with only 5 users

- It's a bit controversial among academics, but it's pretty standard practice in industry
  - More validation that it's effective!

- It doesn't have to be one or the other...
  - Discount usability testing for formative evaluation while the design itself is still change
  - More extensive usability testing for summative evaluation as the application nears completion
    - Can provide evidence for effectiveness and clarify training needs at this stage

Competitive usability testing

- Participants test competing interfaces
  - One version vs. another version of the same interface
  - Interface vs. older version of the same interface
  - Interface vs. a competitor's interface

- We can have **within-subject test designs** where the same participants test both interface, or **between-subject designs** where different participants test the interfaces

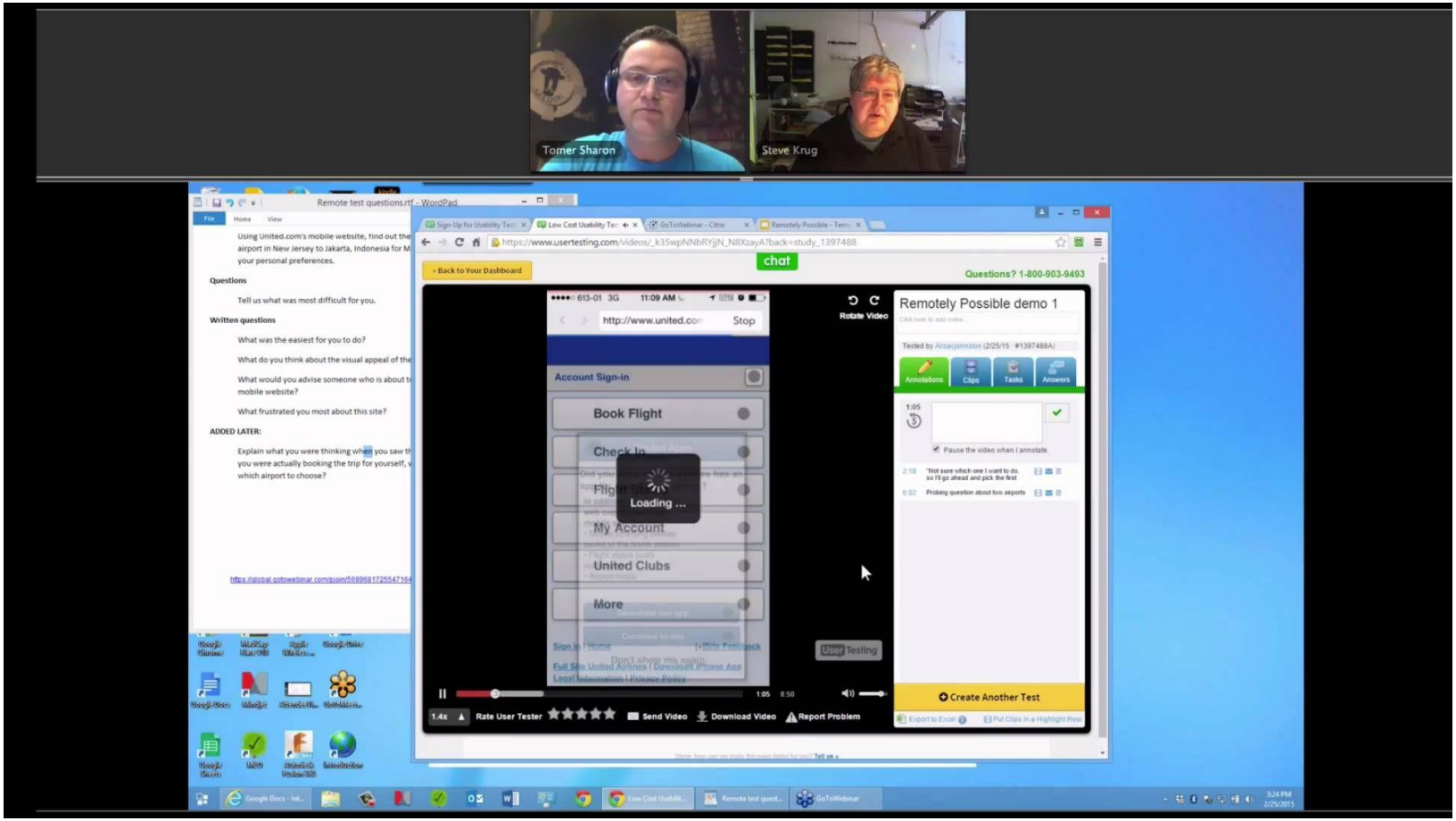
- Within-subjects more typical here, allows participants to make direct comparisons

A/B testing

- Two groups of users are randomly assigned to either a control group (no change) or the treatment group (with the change) and a dependent measure is tested to see if there is a difference between groups
  - The change in A/B testing is some difference in the design of the interface
  - We can use statistical analysis on the results, "with a 99% confidence level, we can say..."
  - Effectively a between-subjects competitive usability test
  - Sometimes A/B testing is considered its own category of testing

- A very widely used form of usability testing in industry
  - e.g. Microsoft has run 200 Bing A/B tests concurrently with 100 million users to test new ideas, modifications

Remote usability testing



Source: https://i.pinimg.com/originals/c3/3e/72/c33e7205dda9fdcbb96927b4aa6d71b6.jpg

Remote usability testing

- Usability testing done online instead of in-person
  - Generally less ability to observe the user's behavior
  - However modern video conference tools (e.g. Teams) have become much better (improved video quality, etc.)

- Can be done **synchronously** with tester observing users as they perform the tasks
  - Sometimes called moderated or monitored remote usability testing

- Or can be done **asynchronously** with users performing tasks and testers observing results later
  - Tester may be able to watch a recorded video
  - Sometimes called remote unmoderated or unmonitored remote usability testing

Remote usability testing

- Can actually be more cost effective than in-person
  - e.g. less equipment is needed
- Can open up test population to a large number of participants, more diverse backgrounds
- Can be more realistic than in-person testing... users use the interface in their real home environment
- Some research has shown remote usability testing can find more usability issues than in-person testing



Source: https://www.forconstructionpros.com/events/event/21062455/webinar-the-fully-automated-foreman

Other types of usability tests

- **Field testing** is focused on testing the interface in realistic or natural environments
  - e.g. testing an interface for a hospital in the hospital
- **Universal usability testing** is focused on testing with diverse users, hardware, software, networks
- **Can-you-break-this testing** is when users are challenged to find fatal flaws in the interface
  - Used frequently with video games

Other types of usability testing

- **Lab testing** is done in a usability lab with equipment such as two-way mirrors and recording devices (cameras, microphones)
  - High-fidelity observations, potentially less realism
  - Sometimes labs resemble a living room or office to add realism to the situation

- **Guerilla testing** is done in a public place with randomly chosen participants
  - e.g. at a coffee shop

Usability testing observation techniques, tools and metrics

Note-taking

- As the users perform tasks with the interface, we can take notes to analyze and summarize later

- Notes can be more formal with categories, checklists and/or questions
  - "How frustrated did the participant appear?"

- But more typically, note-taking is free form written observations of the participant's experience
  - "Appeared confused when trying to find the pay button"
  - "Smiled at the pop sound effect after clicking delete"

Usability metrics

- **Metrics** are the things we measure during a usability test, for example:
  - Time to perform a task (performance)
  - Ability to complete tasks successfully
  - Error rates associated with a task
  - Subjective satisfaction of the user

- **Measures** are how we measure metrics
  - Metrics can be measured in different ways

Usability metrics

- We can measure time to perform tasks and error rates through observation (i.e. a stop watch, a notebook for recording errors)

- **Logging** is when our application records metrics like time to perform tasks and error rates
  - We can add code to our interface to record all user actions, keep track of time, etc.
  - Can be particularly useful for remote usability testing, but can be used anytime
  - High accuracy, can provide exceptional data

Example usability test performance results from Adobe

	Original design	Redesign
Task 1	12 sec.	6 sec.
Task 2	75 sec.	15 sec.
Task 3	9 sec.	8 sec.
Task 4	140 sec.	40 sec.

- Results in the table are average for a set of users
- Original design took 236 seconds total for all tasks
- Redesign took 69 seconds total for all tasks
- We can claim a 242% performance improvement

Source: https://www.nngroup.com/articles/usability-metrics/

Task performance

- Tasks we ask the users to perform could be any tasks or subtasks in a task hierarchy we wish to test
- But typically the tasks that are tested are the most high-level tasks (sometimes called goals), that take more time to perform and are made up of subtasks themselves
- If users are failing to perform tasks, we need to account for this in our data presentation and analysis
  - e.g. look at task performance individually instead, also present a table of percentage failed to perform tasks

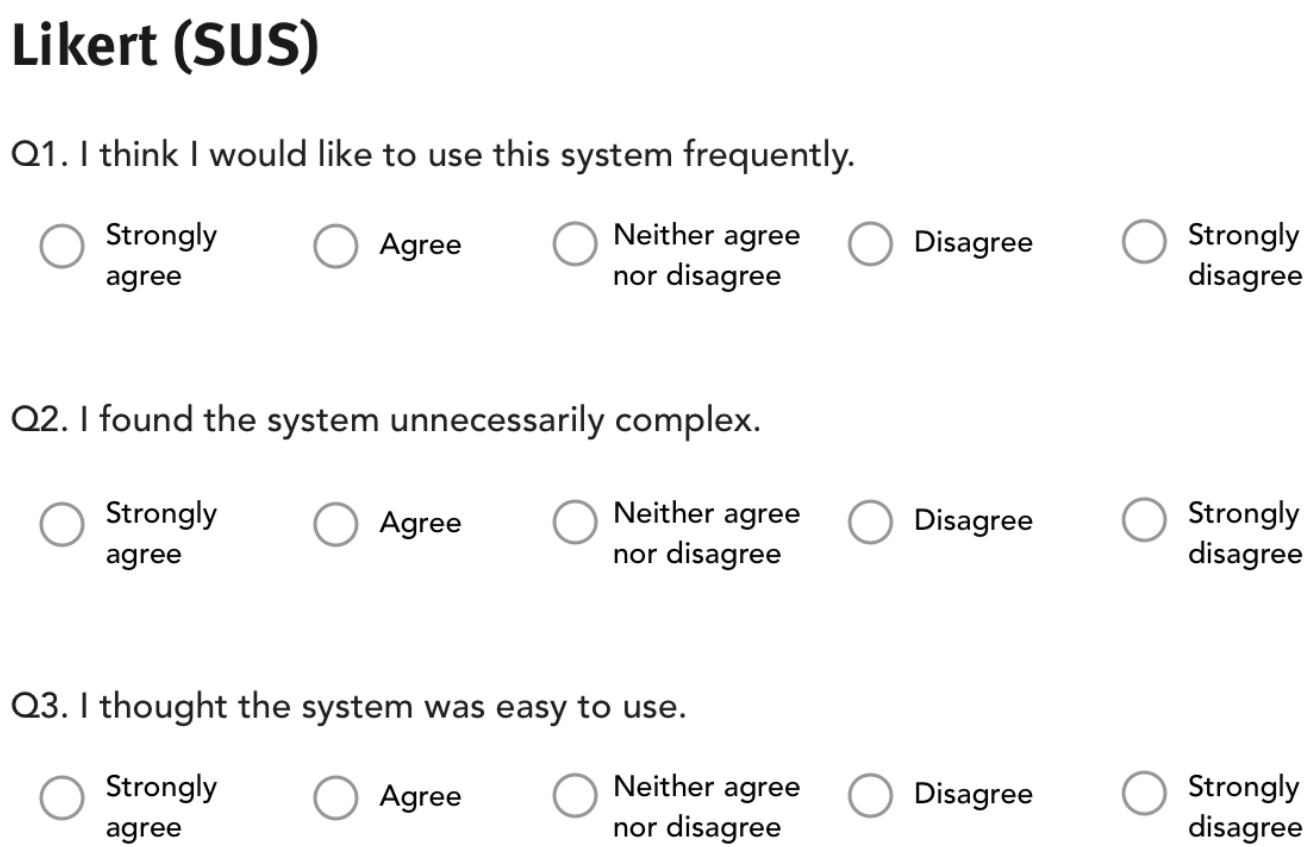
Surveys

- Surveys can be done before and/or after the user uses the interface to collect data as part of a usability test
  - Inexpensive in time and money, modern tools (e.g. Google Forms) make them trivial to prepare
  - Survey data can be anonymous... helps for honesty from participants shy about face-to-face criticism
- Surveys done before testing can be used to collect demographic data and user characteristics
  - e.g. age, gender, education, income, personality, etc.
  - e.g. digital literacy, job responsibilities, familiarity with similar interfaces

Surveys

- Surveys done after testing can be used to collect data about how users think and feel about the UI
- We can use statistical analysis on survey data
  - e.g. is the percentage of the participants that preferred interface X to Y statistically significant within a confidence interval?
- We can use survey data to provide a metric for subjective satisfaction, trustability and other user feelings and thoughts

Likert scale



Source: https://media.nngroup.com/media/editor/2020/05/14/sus_likert.png

Likert scale

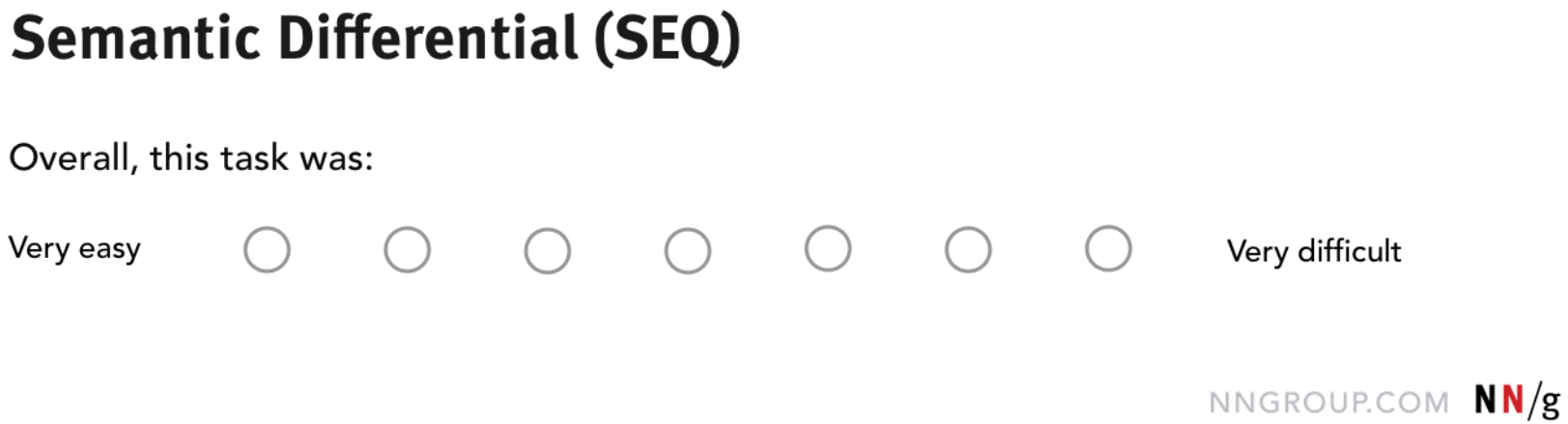
- We can expand the scale to include "somewhat agree", "somewhat disagree", etc.

- We can assign values to each response
  - Strongly disagree – 0
  - Disagree – 1
  - Neutral – 2
  - Agree – 3
  - Strongly agree – 4

- We can then average responses to get a tangible metric for an "agreement level"

Semantic differential

- Can also be used to quantify responses 0,1,2,…
- We can put any "opposites" at the end of each scale, e.g. "fun vs boring", "clear vs confusing"



Source: https://media.nngroup.com/media/editor/2020/05/14/seq_semantic_differential.png

Surveys

- We can ask critical questions about specific aspects of the interface we're concerned about
- We can ask more free form questions that allow users to provide a paragraph of feedback
- We can design the questions for our target user
  - Children, seniors, technical background, etc.

Surveys

- Standardized surveys exist and are commonly used
  - Surveys for measuring usability in general
  - Surveys for measuring specific aspects of usability
  - Surveys for measuring "engagement" and other feelings
- Advantage: a standard measuring stick for usability
  - Can compare one interface to another
- System Usability Scale (SUS)
  - Ten (likert) question survey: https://www.usability.gov/how-to-and-tools/methods/system-usability-scale.html
  - Results can be used to compute score from 0-100

Interviews

- Can also be done before and after usability testing
  - Almost always done at the end of a usability test
  - More expensive in time than surveys
  - Tend to receive less honest feedback than surveys
  - But can ask more probing follow-up questions
- Interview questions can be scripted in advance
  - But interviews can also be done more free form
  - Blend of both is usually best
- Can be good to have two interviewers
  - One to take questions, one to take notes

Conducting interviews

- Decide on a goal(s) for the interview in advance
- Explain the purpose of the interview to the user
  - Helps them to understand how to help you as the interviewer
- Make the user feel comfortable (they will share more openly if they are comfortable)
  - Nod and actively listen to the user
  - Don't rush the user, don't interrupt the user
  - Begin with questions that are easier, less judgmental
  - Be authentic with the user

Conducting interviews

- Prepare questions in advance
  - Create a list, get feedback on the list from peers
  - Ensure that questions aren't **leading questions** that suggest a particular answer
    - e.g. "Would you rather use the old version or this improved version of the application?"
  - Write questions to promote dialog rather than short answers... dialog is advantage of interview vs survey
    - e.g. "did you like X?" is a yes/no short answer question
    - e.g. "what confused you the most about X?" is more likely to prompt a discussion
  - Anticipate responses and create follow-up questions
    - Follow-up questions can be specific, but having a few "go to" follow-up questions is good... "can you tell me more about that?"

Think-alouds

- **Concurrent think aloud (CTA)** is when tester asks participant to think aloud *while using the interface*
  - Goal is to understand participant's stream of thoughts
  - Can't really get accurate usability metrics like performance time

- **Retrospective think aloud (RTA)** is when tester asks participant to retrace their thought process *after* having used the interface
  - Less reliable for understanding participant thoughts
  - Doesn't interfere with usability metrics
  - Increases total test time
  - Can show participants video of themselves using the interface to help with RTA

Probing

- **Concurrent probing** is when the tester asks the participant probing questions about their thoughts as they use the interface
  - e.g. participant becomes confused or excited, tester can ask why in the middle of the test
  - Issue is that this interferes with the testing itself

- **Retrospective probing** is when tester asks probing questions about their thoughts after using the interface
  - Doesn't interfere with testing, but reliability issues
  - Can be done as part of a post-test interview

Other techniques

- **Eye-tracking** software can be used to determine where users look with an interface
  - We can generate "heat maps" from this data
- **First-click testing** where we ask the user where they would first click for starting a task
- **5-second testing** gives users a question, and then shows them the app for 5 seconds to answer it
  - Can users quickly identify something?